

BMEG3105: Data Analytics for Personalized Genomics and Precision Medicine

Lecture 9 - 7 October 2026

Topic : Classification
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1. Previous Lecture Recap

1.1 Classification

Given a collection of records (here the example uses a training set), where each record has a set of attributes and one attribute is the class.

We have to find a method to assign the class of previously unseen records based on their other attributes and the training set accurately.

Person	Height(m)	Weight(kg)	Gender
P1	1.79	75	M
P2	1.64	54	F
P3	1.70	63	M
P4	1.88	78	M
P5	1.75	70	??

1.2 KNN (K-Nearest Neighbor) Classification

The standard procedure of KNN:

- Choose the distance metric and K
- Normalize the data
- Compute the distances
- Identify the K most similar data
- Take out their class and find the mode class

As we chose $K = 2$ and Euclidean distance, we know that from P1 and P2, P5 is also M or Male

1.3 Clustering vs Classification

	Clustering	Classification
Goal	Find similarity (clusters) in the data	Assign class to the new data
Data	Data without class	Training data with class and testing data without class
Classes	Unknown number of classes	Known number of classes
Output	The cluster index for each point	The class assignment of the testing data
Algorithm	One phase	Two phases (training and application)

2. Logistic Regression

Using KNN Classification is one part of the problem. What if we have P1-P10000 as training data? Using this data, we have to store, calculate the distance matrix, and it is very slow to predict. Now we want to use a formula that requires no calculating the distance matrix and to get the results, we just use a simple arithmetic calculation.

2.1 Function

From our observation, if $H + W$ is large, then the person is likely Male. Therefore,

$$H + W \geq 0.5 \rightarrow \text{Male}$$

But how to get the formula and different attributes may not be equally important?

2.2 Adjust the Formula

Now, add weights and bias:

$$\begin{array}{c} H + W \geq 0.5 \rightarrow \text{Male} \\ \downarrow \\ w_h H + w_w W + w_0 \geq 0.5 \end{array}$$

Note that:

- Weights: $w_h H + w_w W$
- Bias: w_0

But what if the weights and bias are large-valued? Therefore, we use the logistic function:

$$\frac{1}{1 + e^{-(w_h H + w_w W + w_0)}} \geq 0.5$$

With:

- $w_h H = 1$
- $w_w W = 1$
- $w_0 = -0.5$

Training : fit the training data to get weights and bias

Testing : run the formula

3. Logistic Regression Model Training

3.1. Training

Training is to get the weights and bias to make the model fit the training data. Other than that, we also want to make

$$\frac{1}{1 + e^{-(w_h H + w_w W + w_0)}} \geq 0.5 \text{ to be correct}$$

$$Y^{output} = \frac{1}{1 + e^{-(w_h H + w_w W + w_0)}} ; \text{ outputs are } \mathbf{1 \text{ for male and } 0 \text{ for female.}}$$

Loss function:

$$(Y^{output} - Y)^2$$

Y = the true label we have for training data

Total loss:

$$L = \sum_{P1}^{P4} (Y^{output} - Y)^2$$

From these functions, we want to **find ws and minimize loss**.

3.2 Find Minimum Value by Calculus

- Find the first derivative that the answer is zero.
- Use second derivative to determine whether it's a maximum, minimum, or an inflexion point

$$\frac{d^2y}{dt^2} < 0 : \text{maximum}$$

$$\frac{d^2y}{dt^2} > 0 : \text{minimum}$$

$$\frac{d^2y}{dt^2} = 0 : \text{inflexion point}$$

3.3 Gradient Descent Algorithm

What if the equation is not easy?

- Initialize weights and bias with random values
- For P1-P4

- Calculate the output Y^{output}

- Update weights

$$w_i = w_i + \Delta w_i$$

$$\Delta w_i = 2 * \alpha (Y - Y^{output}) \frac{\partial Y^{output}}{\partial w_i}$$

α is a small constant

- Repeat until there are no more to update

The best weights $\rightarrow$ make smallest $(Y^{output} - Y)^2$

4. Logistic Regression to Neural Networks

Logistic Regression are:

- Fast prediction
- Successful in real-life
- High tolerance to noisy data
- Long training time
- Poor interpretability

Neural Networks are:

- Layers of neurons
- Complex patterns